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CHE 313 Transport Phenomena III

Exam #1 (10/07/25) **Name:** Quan Huynh

A. (30pts) We have a liquid feed that is 40.0 mol% methane (C1), 20.0 mol% n-butane (C4), 15 .0mol% n-pentane (C5) and unknown mol% of n-hexane (C6). Feed rate (F) is 100 kmol/h and fraction feed vaporized (V/F) is 0.7. Feed pressure is 100.0 psia. The flash drum operates at 250 kPa and 10 °C.

1) (5pts) Sketch the process.

2) (5 pts) Find mole fraction (z_4) of n-hexane in the feed and flow rate of vapor stream (V) and liquid stream (L).

3) (15 pts) Find V/F use Rachford-Rice Equations. $K_{C1} = 50$, $K_{C4} = 0.6$, $K_{C5} = 0.17$, and $K_{C6} = 0.05$. Use $(V/F)_1 = 0.45$ as 1st guess and find V/F value when f(V/f) converges to less than 0.01.

$$f\left[\frac{V}{F}\right] = \sum_{i=1}^{C} \frac{(K_i - 1)z_i}{1 + (K_i - 1)(\frac{V}{F})} = 0$$

$$\left[\frac{V}{F}\right]_{k+1} = \left[\frac{V}{F}\right]_k - \frac{f_k}{\left[\frac{df_k}{d(V/F)}\right]}$$

$$\frac{df_k}{d(V/F)} = -\sum_{i=1}^{C} \frac{(K_i - 1)^2 z_i}{\left[1 + (K_i - 1)(\frac{V}{F})\right]^2}$$

4) (5 pts) Find x_i, y_i, V, and L

$$x_i = \frac{z_i}{1+(K_i-1)\frac{V}{F}} \quad i=1,C \qquad y_i = K_i x_i = \frac{K_i z_i}{1+(K_i-1)\frac{V}{F}} \quad i=1,C$$

5) (5 pts) In the class, we practiced the double trial and error (simultaneous multicomponent convergence) procedures when T_{drum} is unknown. <u>Make your own flowsheet to find T_{drum}, V/F, x_i, y_i, L, and V</u>

B. (70 pts) We are separating acetone and ethanol in a stripping column. The column has two feeds and operates at a pressure of 1.0 atm. Feed 1 is a saturated liquid and is fed into the top of column (no condenser). Flow rate of F1 is 150.0 kmol/hr with 60.0 mol% acetone. Feed 2 is a 10.0 mol% acetone and two-phase mixture with that is 80% vapor. Flow rate of F2 = 50.0 kmol/hr. Assume CMO is valid. Bottoms product is 4.0 mol% acetone and the column has a partial reboiler with bottom up. Boil-up ratio ($\overline{V}/B$) is 1.5.

1) (5 pts) Draw a schematic diagram of the distillation column and label all known variables (use V & L for enriching section and $\overline{L}$ & $\overline{V}$ for stripping section)

2) (5 pts) There are total three balance envelopes required for this question (enriching & Feed 1 section, stripping section, and Feed2). Draw separately from (a) and write down mass balance for each section.

3) (5 pts) Set up overall mass balance and component mass balance

4) (10 pts) Find L, $\overline{L}$, B, $\overline{V}$, V, and D

5) (10 pts) The equation for the feed line is:

$$y = \frac{q_i}{q_i - 1}x + \frac{z_i}{1 - q_i}$$

$$q_i = \frac{L_{Fi}}{F_i}$$

Find q_1, q_2, coordinates on y=x line, and slopes of two feed lines (F1 & F2), and y-intercept of F2.

6) (5 pts) Find mole composition of ethanol in distillate stream (y_D)

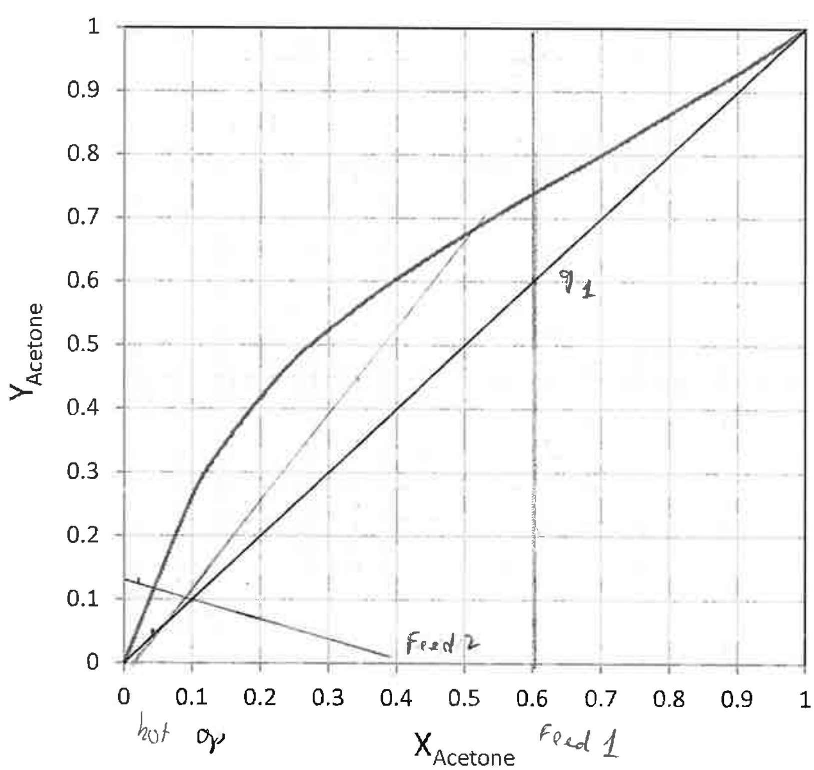
7) (5 pts) Set up component mass balance equation for the top & feed 1 section using mass balance from (b). Then, derive operating equation for the top section.

8) (5 pts) Find the slope, y-intercept, and the intersection point of Feed 1 and top operating line.

9) (5 pts) Find slope and coordinate on y=x curve of the bottom operating line.

$$y = \left(\frac{\bar{L}}{\bar{V}}\right)x - \left(\frac{\bar{L}}{\bar{V}} - 1\right)x_B$$

10) (10 pts) Draw the top operating, bottom operating, two feeds, and middle operating lines



11) (5 pts) Find the number of stages and the optimum locations of feed 1 and feed 2 stages. Step off stage from the bottom.

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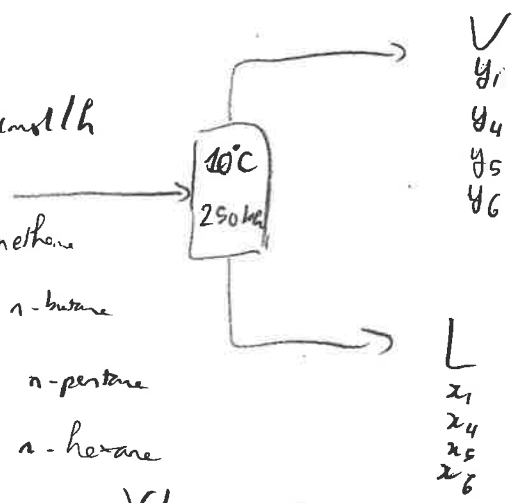
Part A

1) $F = 100\ kmol/h$



$z_{C1} = 0.4$ methane

$z_{C4} = 0.2$ n-butane

$z_{C5} = 0.15$ n-pentane

$z_{C6} = ?$ n-hexane

$V/F = 0.7$

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2) $V/F = 0.7 \Rightarrow V = 0.7F = 0.7 \times 100 = \boxed{70\ kmol/h}$

$V + L = F \Rightarrow L = F - V = 100 - 70 = 30\ kmol/h$

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$z_1 + z_4 + z_5 + z_6 = 1 \Rightarrow 0.4 + 0.2 + 0.15 + z_6 = 1 \Rightarrow \boxed{z_{C6} = 0.25}$

3)

$$f(V/F) = \sum_{i=1}^{4} \frac{(k_i - 1)z_i}{1 + (k_i - 1)\frac{V}{F}} = 0$$

$$f\left(\frac{V}{F}\right) = \frac{(k_{C1} - 1)z_{C1}}{1 + (k_{C1} - 1)V/F} + \frac{(k_{C4} - 1)z_{C4}}{1 + (k_{C4} - 1)V/F} + \frac{(k_{C5} - 1)z_{C5}}{1 + (k_{C5} - 1)V/F} + \frac{(k_{C6} - 1)z_{C6}}{1 + (k_{C6} - 1)V/F} = 0$$

$$= \frac{(50 - 1)0.4}{1 + (50 - 1)V/F} + \frac{(0.6 - 1)\cancel{0.25}\ 0.2}{1 + (0.6 - 1)V/F} + \frac{(0.17 - 1)0.15}{1 + (0.17 - 1)V/F} + \frac{(0.05 - 1)0.25}{1 + (0.05 - 1)V/F}$$

$$= \frac{19.6}{1 + 49V/F} - \frac{0.1}{1 - 0.4V/F} - \frac{0.1245}{1 - 0.83V/F} - \frac{0.2375}{1 - 0.95V/F}$$

$$f\left(\frac{V}{F}=0.45\right) = \frac{19.6}{1+49\times0.45} - \frac{0.1}{1-0.4\times0.45} - \frac{0.1245}{1-0.83\times0.45} - \frac{0.2375}{1-0.95\times0.45} = 0.115$$

$\Rightarrow$ need more itterations

$$\frac{df}{d(V/F)} = \frac{-960.4}{(1+49V/F)^2} - \frac{0.04}{(1-0.4V/F)^2} - \frac{0.1034}{(1-0.87V/F)^2} - \frac{0.2256}{(1-0.95V/F)^2}$$

method is ok!

$$\left(\frac{V}{F}\right)_2 = \left(\frac{V}{F}\right)_1 - \frac{f_1}{\frac{df_1}{d(V/F)}} = 0.45 + \frac{0.115}{2.819} = 0.4907$$

$f\left(\frac{V}{F} = 0.4905\right) = 0.0031 < 0.01$

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$\Rightarrow \frac{V}{F} = \boxed{0.4909}$

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4) $x_{C1} = \frac{z_{C1}}{1+(k_{C1}-1)\frac{V}{F}} = \frac{0.4}{1+(50-1)0.4907} = \boxed{0.0160} \Rightarrow y_{C1} = 50x_{C1} = \boxed{0.7986}$

$x_{C4} = \frac{0.2}{1+(0.6-1)0.4907} = \boxed{0.2488} \Rightarrow y_{C4} = 0.6\cdot x_{C4} = \boxed{0.1493}$

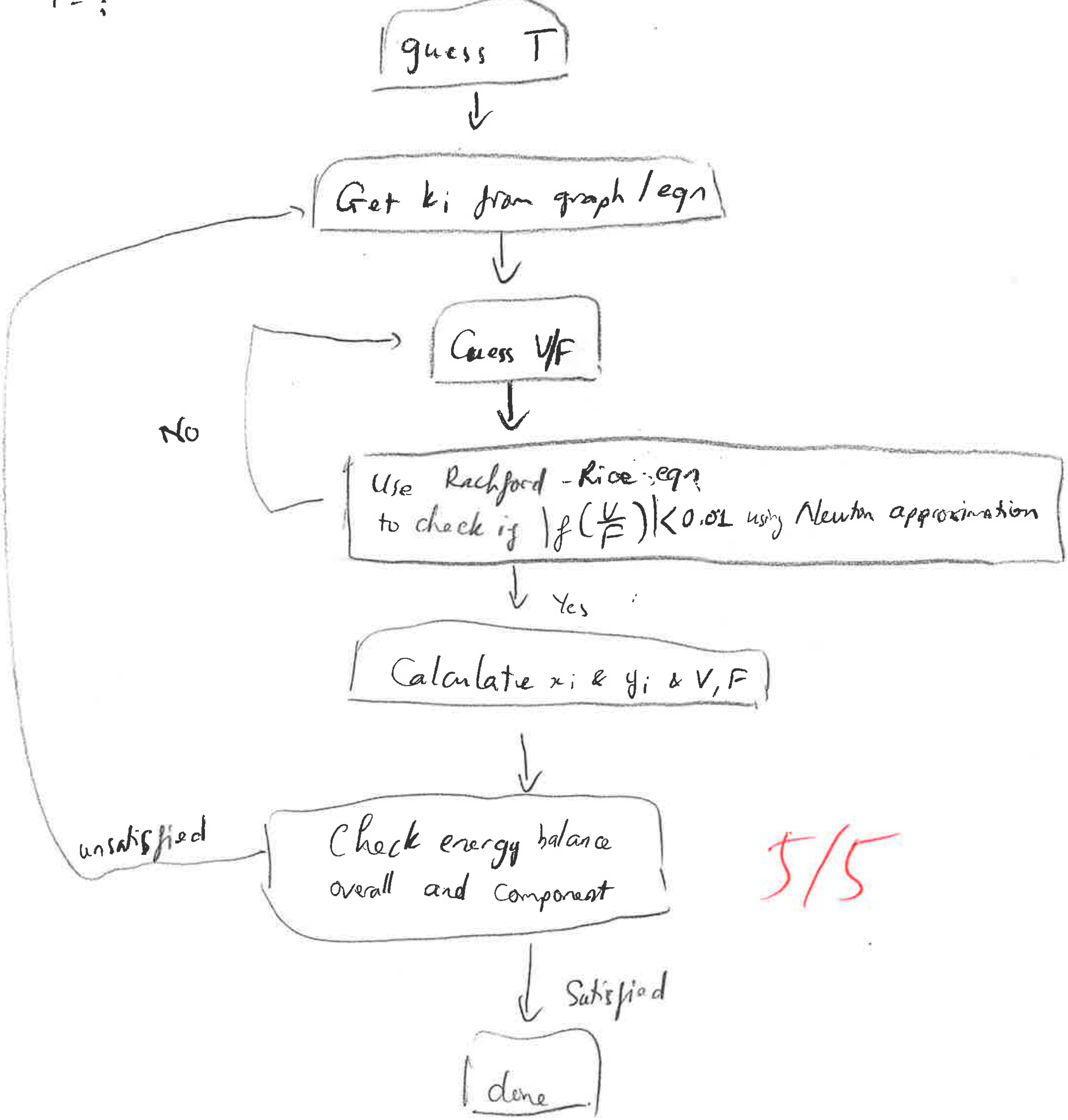
$x_{C5} = \frac{0.15}{1+(0.17-1)0.4907} = \boxed{0.2531} \Rightarrow y_{C5} = 0.17\, x_{C5} = \boxed{0.0430}$

$x_{C6} = \frac{0.25}{1+(0.05-1)0.4907} = \boxed{0.4683} \Rightarrow y_{C6} = 0.05\, x_{C6} = \boxed{0.0234}$

$\frac{V}{F} = 0.4907 \Rightarrow V = 0.4909F = \boxed{49.07\ kmol/h}$

$L = F - V = 100 - 49.09 = \boxed{50.93\ kmol/h}$

5) $T = ?$


guess T
Get k_i from graph / eqn
Guess V/F
Use Rachford - Rice eqn to check if $|f(\frac{V}{F})| < 0.01$ using Newton approximation
No
Yes
Calculate x_i & y_i & V, F
Check energy balance overall and Component
unsatisfied
Satisfied
done

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$z_1 = 0.6$

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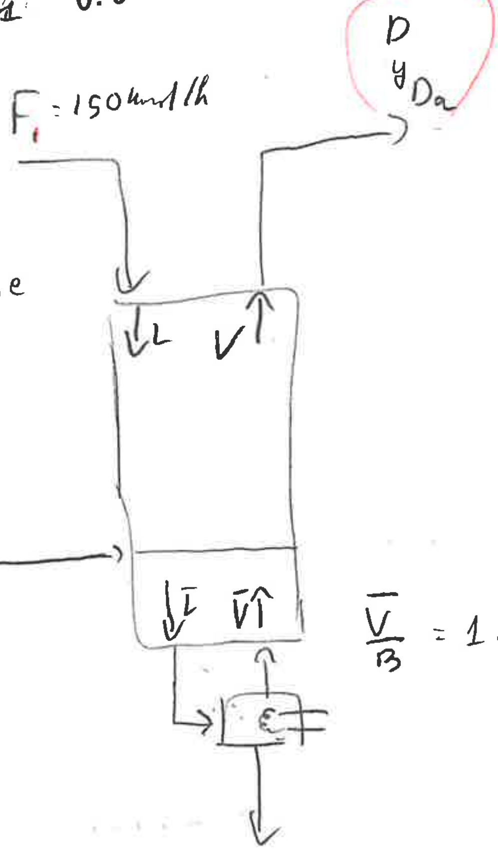
Part B

1)



Sat. liq

Target component: acetone

$F_2 = 50$ kmol/h

$z_2 = 0.1$

2phase, 80% vap

$\frac{\bar{V}}{B} = 1.5$

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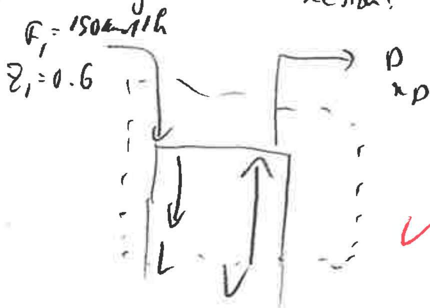
B

$x_B = 0.04$

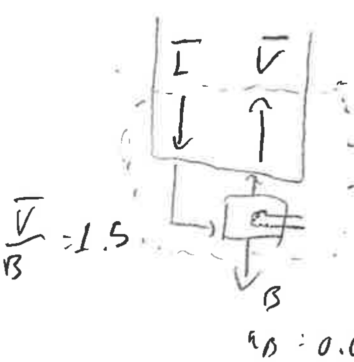
2) enriching & Feed 1 section:



M.B:

$$\begin{cases} F_1 + V = L + D \\ F_1 z_1 + V y = L x + D x_D \end{cases}$$

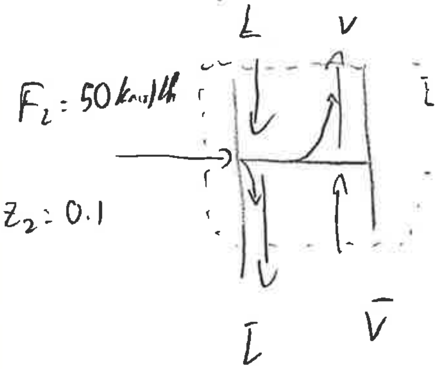
Stripping section:



M.B:

$$\begin{cases} \bar{L} = \bar{V} + B \\ \bar{L} x = \bar{V} y + B x_B \end{cases}$$

Feed 2:



M.B:

$$\begin{cases} F_2 + L + \bar{V} = \bar{L} + V \\ F_2 z_2 + Lx + \bar{V}y = \bar{L}x + Vy \end{cases}$$

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3) System M.B:

$$\begin{cases} F_1 + F_2 = B + D \\ F_1 z_1 + F_2 z_2 = B x_B + D y_D \end{cases}$$

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4) 0/10

4) Stripping section:

$$\begin{cases} \bar{L} = \bar{V} + B \\ \bar{L}x = \bar{V}y + Bx_B \Rightarrow y = \frac{\bar{L}}{\bar{V}}x - \frac{B}{\bar{V}}x_B \end{cases}$$

$\frac{\bar{V}}{B} = 1.5 \Rightarrow \frac{B}{\bar{V}} = \frac{2}{3}$

$\frac{\bar{L}}{\bar{V}} = \frac{\bar{V}+B}{\bar{V}} = \frac{\frac{\bar{V}}{B}+1}{\frac{\bar{V}}{B}} = \frac{1.5+1}{1.5} = 5/3$

$\Rightarrow$ Stripping op line: $y = \frac{5}{3}x - \frac{2}{3} \cdot 0.04$

y-intersect: $y=0 \Rightarrow x = 0.016$ slope $= 5/3$

$\Rightarrow$ pass thru $(0.04, 0.04)$ and $(0.016, 0)$

9) $y = \frac{\bar{L}}{\bar{V}}x - \left(\frac{\bar{L}}{\bar{V}} - 1\right)x_B$

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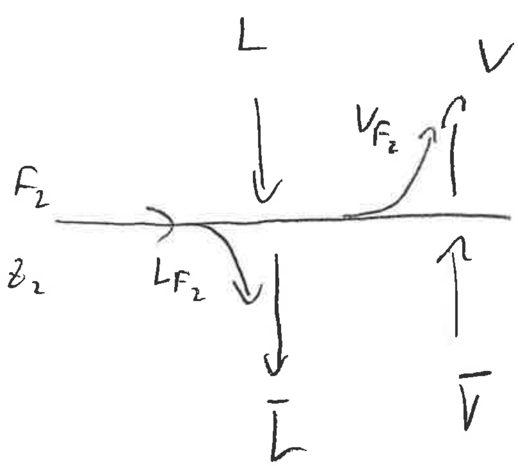
$\frac{\bar{L}}{\bar{V}} = \frac{5}{3}$, pass thru $(0.04, 0.04)$

slope: $\bar{L}/\bar{V} = 1.67$

$(0.61, 1)$

5)



$L + L_{F_2} = \bar{L}$ $\qquad$ $\bar{V} + V_{F_2} = V$

given : $\frac{V_{F_2}}{F_2} \overset{\nearrow f_2}{=} 0.8 \Rightarrow \frac{L_{F_2}}{F_2} \overset{q_2}{=} 0.2$

$y = \cancel{0.8} \; \frac{0.2}{0.2-1} x + \frac{0.1}{1-0.2} \Rightarrow y = \frac{-0.2}{0.8} x + \frac{0.1}{0.8}$

$y = \frac{-1}{4} x + \frac{1}{8}$

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Passing thru $y = x = z_2 = 0.1$

Top op line : $y = \frac{L}{V} x + \frac{D x_D - F_1 z_1}{V}$

5)

$= 1.1x + 0.018.$

$y_1 = \frac{q_1}{q_1 - 1} x + \frac{z_1}{1-q_1}$

$q_1 = 1$ (sat. liq) $\Rightarrow$ slope = $\frac{1}{\infty}$ $\Rightarrow$ op. eqn for F_1 : $x = z_1 = 0.6$

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(0) 4/10